Balanced Expected-Pyramid KVPress Compression for Training-Free Pythia-70M Inference

Group Project Team
Shanghai Jiao Tong University
https://github.com/SJTU-huaxing/llm-hw-team.git

Abstract

KV cache compression reduces autoregressive inference memory, but aggressive pruning often trades quality for speed. We study this tradeoff on EleutherAI/Pythia-70M with NVIDIA KVPress, no training, and fp32 eager attention. Beyond Dense, StreamingLLM, SnapKV, PyramidKV, ExpectedAttention, ObservedAttention, and TOVA baselines, we iteratively test hybrid and layer-budget variants. The selected aggressive-compression method is Expected Soft Pyramid: it keeps ExpectedAttention as the token scorer and applies a mild low-to-high layer budget schedule. At a 132-token target cache for a 512-token context, it lies on the Pareto frontier and has the best equal-rank balance over average perplexity, TPOT, and decode-attention FLOPs. We also report that adding observed-attention residuals often hurts WikiText quality, which makes the conservative ExpectedAttention-based variant the most reliable innovation.

1 Introduction

The KV cache stores one key and value vector per layer and token, so long-context decoding stresses GPU memory and attention bandwidth. The assignment asks us to combine ideas from four levels: attention mechanisms, per-layer KV eviction, inter-layer cache allocation, and macro inference architecture. We keep the model fixed, use no training, and use KVPress [NVIDIA, 2025] to hook Transformers during prefill.

Pythia-70M [Biderman et al., 2023] uses GPT-NeoX attention modules and partial RoPE, while several KVPress utilities assume LLaMA-like module names and full-dimensional RoPE. Our adapter only supplies these structural aliases and the partial-RoPE correction needed for ExpectedAttention. All compression decisions are implemented as KVPress-style presses that gather existing KV tensors along the sequence dimension.

2 Method

Table 1 summarizes how the implementation covers the assignment layers. Official baselines include attention sink retention [Xiao et al., 2023], observed attention from SnapKV/PyramidKV [Li et al., 2024, Cai et al., 2024], H2O-style heavy-hitter scoring [Zhang et al., 2023], and TOVA [Oren et al., 2024].

Expected Soft Pyramid uses the ExpectedAttention score from KVPress but changes the budget per layer:

$$B_\ell = \mathrm{round}(B\alpha_\ell), \quad \alpha_\ell \in [1.25, 0.75],$$

where α_ℓ decreases linearly from lower to higher layers and averages to one. This keeps the global target cache size unchanged while testing the common observation that lower layers can benefit from

Preprint.

Table 1: Mapping from acceleration layers to implemented components.

Layer	Components
Attention mechanism	ExpectedAttention, ObservedAttention/TOVA, value norm
Per-layer compression	StreamingLLM, SnapKV, PyramidKV, observed/top-k scorers
Inter-layer compression	Pyramid budgets, lazy gate, expected soft budget ablation
Macro architecture	KVPress pipeline, prefill-once compression, unified timing

broader lexical context. Sink tokens are always protected. We also test Hybrid Expected-Observed,

$$s = 0.55\,\hat{s}_{\text{expected}} + 0.30\,\hat{s}_{\text{observed}} + 0.15\,\hat{s}_{\text{vnorm}},$$

where each score is min-max normalized per batch and head. Sink tokens and a 32-token recent window receive a protection boost before top-k selection. Residual and soft-pyramid hybrid variants are retained as ablations.

3 Experiments

We evaluate 'EleutherAI/pythia-70m' on WikiText-2 test [Merity et al., 2017] and one PG-19 test stream sample [Rae et al., 2020]. For PPL, a 512-token context is prefetched and compressed before scoring the next 128 continuation tokens. For generation, a 512-token PG-19 prompt generates 32 tokens. The decode attention FLOPs proxy is $\sum_{\ell,t} 4HDL_{\ell,t}$. Reported runs use an NVIDIA GeForce RTX 5060 Laptop GPU.

We first screen new baselines and hybrids on 256-token contexts and target cache sizes 96, 128, and 192. The strongest new official baselines are ObservedAttention and TOVA, so the final table keeps them alongside the original methods. The balanced selection rule filters methods with kept fraction above 0.5 and ranks the remaining methods by average PPL, TPOT, and FLOPs ratio with equal weight.

Table 2: Target cache 132. PPL is averaged over WikiText-2 and PG-19; generation is on PG-19.

Policy	Avg PPL	Kept	TPOT	FLOPs
Dense	21.344	1.000	0.006071	1.000
ExpectedAttention	26.326	0.258	0.006373	0.281
Expected Soft Pyramid	28.342	0.258	0.006066	0.281
Expected+Pyramid	35.066	0.259	0.006056	0.282
Hybrid Expected-Observed	37.870	0.258	0.006080	0.281
Hybrid Soft Pyramid	38.884	0.258	0.006064	0.281
SnapKV	42.340	0.258	0.005959	0.281
StreamingLLM	46.148	0.258	0.006148	0.281
TOVA	53.386	0.258	0.006146	0.281

Table 3: Balanced selections across target cache sizes.

Target	Selected policy	Avg PPL	TPOT	FLOPs
132	Expected Soft Pyramid	28.342	0.006066	0.281
192	ExpectedAttention	22.407	0.006006	0.395
256	Hybrid Expected-Observed	26.329	0.006105	0.516

4 Analysis and Limitations

The main result is not that a more complicated scorer always wins. ExpectedAttention is clearly the quality anchor, especially on WikiText-2. The successful innovation is conservative: Expected Soft Pyramid changes the inter-layer budget while keeping the ExpectedAttention scorer intact. It gives up about 2 average PPL relative to ExpectedAttention at target 132, but improves TPOT and remains on the Pareto frontier. In contrast, adding observed-attention residuals improves some PG-19 cases

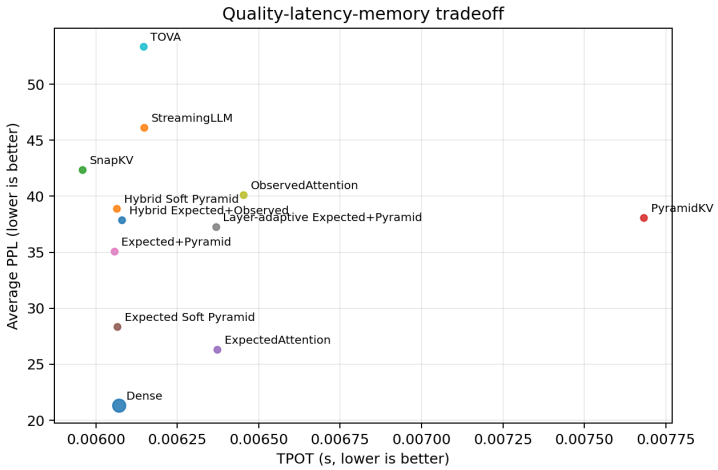

Figure 1: Average PPL versus TPOT at target cache 132.

but tends to damage WikiText quality, so we report those variants as negative ablations rather than the recommended method.

The evaluation is intentionally small and reproducible: one model, no training, two datasets, fixed seeds, and one GPU. Timing differences are small at Pythia-70M scale, so the paper reports exact CSVs rather than over-claiming hardware speedups. We do not implement MQA/GQA/MLA or YOCO-style cross-layer sharing because those require architecture or weight changes outside the no-training setting. KVPress SimLayerKV is excluded because its full-RoPE assumption is incompatible with GPT-NeoX partial RoPE without further changes.

5 Team Contributions

Replace these placeholders with final member names and IDs before submission. Member A: method design and implementation (40%). Member B: experiments and analysis (35%). Member C: paper writing and reproducibility (25%).

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